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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
J ShapeDemo.java > ShapeDemo
1  class Shape {
2      void area() {
3          System.out.println(x: "Area Calculation");
4      }
5  }
6
7  class Circle extends Shape {
8      double radius = 5;
9
10     void area() {
11         double a = 3.14 * radius * radius;
12         System.out.println("Area of Circle = " + a);
13     }
14 }
15
16 class Rectangle extends Shape {
17     int length = 10, width = 5;
18
19     void area() {
20         int a = length * width;
21         System.out.println("Area of Rectangle = " + a);
22     }
23 }
24
25 public class ShapeDemo {
26     Run | Debug
27     public static void main(String[] args) {
28         Circle c = new Circle();
29         Rectangle r = new Rectangle();
30
31         c.area();
32         r.area();
33     }
34 }
```

OUTPUT:

```
[Running] cd "c:\Users\Arpit\java_assignments\" && javac ShapeDemo.java && java ShapeDemo
Area of Circle = 78.5
Area of Rectangle = 50

[Done] exited with code=0 in 1.411 seconds
```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
J Ecommerce.java > Ecommerce
1  interface Product {
2      void display();
3  }
4
5  class ProductDetails {
6      String name;
7      double price;
8
9      ProductDetails(String name, double price) {
10         this.name = name;
11         this.price = price;
12     }
13 }
14
15 class Electronic extends ProductDetails implements Product {
16     Electronic(String name, double price) {
17         super(name, price);
18     }
19
20     public void display() {
21         System.out.println(x: "Electronic Product");
22         System.out.println("Name: " + name);
23         System.out.println("Price: " + price);
24     }
25 }
26
27 class Clothing extends ProductDetails implements Product {
28     Clothing(String name, double price) {
29         super(name, price);
30     }
31
32     public void display() {
33         System.out.println(x: "Clothing Product");
34         System.out.println("Name: " + name);
35         System.out.println("Price: " + price);
36     }
37 }
```

```

38
39 class Grocery extends ProductDetails implements Product {
40     Grocery(String name, double price) {
41         super(name, price);
42     }
43
44     public void display() {
45         System.out.println(x: "Grocery Product");
46         System.out.println("Name: " + name);
47         System.out.println("Price: " + price);
48     }
49 }
50
51 public class Ecommerce {
52     Run | Debug
53     public static void main(String[] args) {
54
55         Electronic e = new Electronic(name: "Laptop", price: 50000);
56         Clothing c = new Clothing(name: "T-Shirt", price: 800);
57         Grocery g = new Grocery(name: "Rice", price: 1200);
58
59         e.display();
60         System.out.println();
61
62         c.display();
63         System.out.println();
64
65         g.display();
66     }
67

```

OUTPUT:

```

[Running] cd "c:\Users\Arpit\java_assignments\" && javac Ecommerce.java && java Ecommerce
Electronic Product
Name: Laptop
Price: 50000.0

Clothing Product
Name: T-Shirt
Price: 800.0

Grocery Product
Name: Rice
Price: 1200.0

[Done] exited with code=0 in 1.425 seconds

```